

MFG-01: Carrington Storm Motors — Manufacturing Strategy

Document ID: CSM/MFG/01 **Version:** 1.0 — Investor-Grade **Classification:** Confidential **Heuristic Foundation:** El Segundo (phased scale-up), Accountant (cost discipline), Zirconia (process engineering) **Agents Consulted:** Manufacturing domain agent, Zirconia, Accountant, El Segundo **Date:** July 2026

1.0 EXECUTIVE SUMMARY

Carrington Storm Motors’ manufacturing strategy follows a phased approach from lab-scale production through pilot facility, volume manufacturing, and ultimately a global manufacturing network. The strategy balances the urgency of deploying grid protection (the world is vulnerable NOW) with the discipline of manufacturing scale-up (rushing causes quality failures that destroy credibility).

The four-phase approach: - **Phase 0 (Current):** Lab-scale production at 1-10 unit batches. R&D focus, not production focus. - **Phase 1 (Year 1-2):** Pilot manufacturing facility producing 100-1,000 units/year. Process development, quality system implementation, supply chain establishment. - **Phase 2 (Year 3-4):** Volume manufacturing plant at 10,000-100,000 units/year. Automation, cost reduction, multi-shift operations. - **Phase 3 (Year 5+):** Global manufacturing network at 1M+ units/year. Regional facilities, licensed manufacturing partners, technology transfer.

As El Segundo frames it: “The grid doesn’t have time for us to spend five years perfecting a manufacturing process. But a failed product deployed at scale destroys more than it protects. Walk the line between speed and quality. That’s the manufacturing challenge.”

2.0 PHASE 0 — LAB-SCALE PRODUCTION (CURRENT)

2.1 Current Capabilities

Process	Scale	Output	Limitations
MXene synthesis	1-10g batches	Lab samples for characterization	Manual operation; batch-to-batch variation; not scalable
MXene film formation	Hand-casting, 10-100 cm ²	Test coupons for TEM cell	Manual; thickness inconsistency
BFRP composite forming	Hand lay-up, small coupons	Mechanical test specimens	Manual; not production-representative
CNT dispersion	Lab-scale sonication, <1L	Conductive composite samples	Manual; inconsistent dispersion
Ceramic processing	Lab-scale ball milling, pressing	Dielectric test pellets	Not representative of production properties
Geopolymer mixing	Lab mixer, small batches	Test cubes/cylinders	Manual; mix consistency variable

2.2 Phase 0 Objectives (Not Manufacturing — R&D)

- Validate material properties (complete — 12 CSMQuantum volumes)
 - Establish process parameters (in progress)
 - Determine scalability constraints (in progress)
 - Design pilot-scale equipment specifications (in progress)
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3.0 PHASE 1 — PILOT FACILITY (YEAR 1-2)

3.1 Pilot Facility Specifications

Parameter	Specification
Facility size	10,000-30,000 sq ft
Clean room	Class 10,000 (ISO 7) for MXene processing (air-sensitive materials)
Production capacity	100-1,000 units/year (Aegis Home equivalent)
Staffing	15-30 personnel (engineers, technicians, quality, supply chain)
Capital investment	\$5-10M (equipment + facility buildout)
Timeline to operational	12-18 months from funding

3.2 Pilot Production Lines

Line	Products	Key Process Equipment
MXene Line	MXene film and coating for Aegis, Charlemagne components	Pilot reactor (10-100L); continuous casting/drying line; roll-to-roll coater
Composite Line	BFRP structural components	Pultrusion line; compression molding press; curing ovens
CNT Line	Conductive composites for grounding systems	CNT dispersion system; injection molding
Ceramic Line	Dielectric components	Tape caster; sintering furnace
Geopolymer Line	Structural pads, enclosures	Mixing/batching system; casting forms
Assembly Line	Final product assembly	Semi-automated assembly stations; testing stations

3.3 Pilot Objectives

Objective	Success Metric
Demonstrate scalable MXene production	Batch-to-batch property variation <10% at 1kg batch size
Validate manufacturing cost model	Actual cost within 20% of estimated cost
Produce first 100+ commercial units	Units pass final acceptance testing
Train manufacturing workforce	Core team proficient in all production processes
Establish quality management system	ISO 9001-ready documentation and processes
Qualify critical suppliers	3+ qualified suppliers for each critical material

4.0 PHASE 2 — VOLUME MANUFACTURING (YEAR 3-4)

4.1 Volume Facility Specifications

Parameter	Specification
Facility size	100,000-250,000 sq ft
Clean room	Class 10,000-100,000 (ISO 7-8) depending on zone
Production capacity	10,000-100,000 units/year
Staffing	100-300 personnel
Capital investment	\$25-75M (equipment + facility)
Timeline to operational	18-24 months from funding decision

4.2 Automation Strategy

Process	Phase 1 (Pilot)	Phase 2 (Volume)	Automation Investment
MXene synthesis	Manual with automated controls	Semi-automated with SCADA	\$2-5M
MXene film formation	Semi-automated roll-to-roll	Fully automated production line	\$3-8M
BFRP pultrusion	Manual feed, automated pull	Fully automated from creel to cutoff	\$2-5M
Assembly	Manual with fixtures	Semi-automated with robotic assistance	\$3-8M
Testing	Manual sample testing	In-line automated testing	\$2-5M
Packaging	Manual	Semi-automated	\$1-3M

4.3 Workforce Plan

Role Category	Phase 1 Headcount	Phase 2 Headcount	Phase 3 Headcount
Process Engineers	3-5	10-20	30-60
Manufacturing Engineers	2-4	10-20	30-60
Quality Engineers / Technicians	2-4	8-15	25-50
Production Operators / Technicians	8-15	50-150	300-600
Maintenance Technicians	1-3	8-15	30-60
Supply Chain / Logistics	1-3	5-15	20-50
Production Management	1-2	5-10	15-30
EHS (Environmental, Health, Safety)	0-1	2-5	8-15
TOTAL	18-38	100-250	460-935

5.0 PHASE 3 — GLOBAL MANUFACTURING NETWORK (YEAR 5+)

5.1 Global Facility Strategy

Region	Facility Type	Timeline	Rationale
United States (Region 1 — East)	Primary manufacturing + HQ	Year 3-4	Domestic production; government contract preference
United States (Region 2 — West)	Secondary manufacturing	Year 5-6	West Coast market; Asia export hub
European Union	Regional manufacturing + sales	Year 5-7	EU market access; local production for EU regulatory preference
Asia-Pacific	Regional manufacturing	Year 6-8	Asia-Pacific market; supply chain proximity (basalt fiber, CNTs)
Canada	Regional manufacturing	Year 4-6	High GIC exposure; USMCA trade benefits; allied nation
Latin America	Licensed manufacturing	Year 7-9	Developing market; local content requirements

5.2 Make vs. Buy Analysis

Component	Make (In-House)	Buy (Outsource)	Rationale
MXene synthesis	YES	NO	Core IP; trade secret process
MXene film/coating	YES	NO	Core IP; quality-critical
BFRP pultruded profiles	YES (Phase 1-2); BUY (Phase 3 from qualified suppliers)	YES (mature technology; commoditizing)	IP in the composite DESIGN, not the pultrusion process
CNT purchase	NO	YES	CNT production is capital-intensive; better to buy from established producers
CNT dispersion/compounding	YES	NO	Dispersion quality is IP-critical
Ceramic dielectric components	YES	NO	Formulation and processing are IP-critical
Geopolymer concrete	YES	YES (from local ready-mix partner)	IP in mix design; local production reduces shipping
Electronics (monitoring, sensors)	NO	YES	Commodity electronics; contract manufacturing
Enclosures / housings	NO	YES	Sheet metal fabrication; contract manufacturing
Fasteners, connectors, cables	NO	YES	Commodity components
Packaging	NO	YES	Commodity

6.0 ERP/MES SYSTEMS

6.1 System Architecture

System	Purpose	Phase 1	Phase 2	Phase 3
ERP (Enterprise Resource Planning)	Financial, procurement, inventory, order management	Oracle NetSuite	NetSuite or SAP S/4HANA	SAP S/4HANA
MES (Manufacturing Execution System)	Shop floor control, production tracking, quality data	Manual + spreadsheets	MES integrated with ERP	Enterprise MES
PLM (Product Lifecycle Management)	Product design, BOM, change management	Document-based	Arena or Propel PLM	Enterprise PLM
QMS (Quality Management System)	Document control, CAPA, audits, training	Cloud QMS (Greenlight Guru, Qualio)	Enterprise QMS (MasterControl, Veeva)	Enterprise QMS

System	Purpose	Phase 1	Phase 2	Phase 3
SCADA (Supervisory Control and Data Acquisition)	Process control and monitoring	Basic PLC controls	Integrated SCADA	Enterprise SCADA across all facilities
WMS (Warehouse Management System)	Inventory, picking, shipping	ERP-native WMS	Dedicated WMS	Enterprise WMS

7.0 MANUFACTURING LOCATION SELECTION

7.1 Key Selection Criteria

Criteria	Weight	Notes
Workforce availability and skills	25%	Manufacturing engineers, chemical process operators, composite technicians
Energy cost and reliability	15%	High-temperature processes (sintering, pultrusion) are energy-intensive
Logistics infrastructure	15%	Raw material delivery and finished product shipping
State/local incentives	10%	Tax abatements, training grants, infrastructure support
Proximity to research partners	10%	University collaborations; talent pipeline
Cost of living / labor costs	10%	Affects both direct labor cost and employee attraction/retention
Regulatory environment	10%	Permitting timeline; environmental regulations; labor laws
Quality of life	5%	Employee attraction in competitive labor market

7.2 Short-List Candidate Regions

Region	Strengths	Concerns
Tennessee Valley (TN, AL, GA)	TVA electricity (low cost, reliable); manufacturing workforce; automotive supplier ecosystem; ORNL proximity	Tornado risk
Upper Midwest (MN, WI, MI)	Strong manufacturing heritage; engineering workforce; high GIC exposure (test environment)	Winter logistics; higher energy costs
Texas Triangle (DFW-Houston-SA-Austin)	Business-friendly; central logistics; large workforce; low energy costs	ERCOT grid reliability concerns (ironic for a grid resilience company)
Carolina Piedmont (NC, SC)	Manufacturing workforce; research triangle universities; port access	Hurricane risk
Ohio River Valley (OH, IN, KY)	Manufacturing corridor; logistics; reasonable costs	Workforce competition with resurgent manufacturing

8.0 AGENT VOICES

Manufacturing Domain Agent: “Manufacturing MXene at scale has never been done. The process is chemically aggressive (fluoride etchants), water-intensive, and sensitive to processing conditions. The pilot facility is not just a step toward volume — it’s a proof of manufacturability. Until we produce 100+ units with consistent properties, CSM is a materials science project, not a manufacturing company. The pilot is the proving ground.”

Zirconia (Process Quality): “The difference between lab-scale MXene and production-scale MXene is the difference between a Michelin-starred meal and McDonald’s. Both are food. Only one is consistent across millions of servings. CSM’s manufacturing challenge is to make MXene not like a Michelin meal (exquisite but inconsistent) and not like McDonald’s (consistent but mediocre) — but like a great regional chain: consistently excellent at scale. That’s the process engineering challenge.”

The Accountant (Cost Discipline): “Manufacturing cost reduction follows a learning curve: 15-25% reduction in unit cost per doubling of cumulative volume. The first 100 units will be expensive — perhaps 5-10x the target unit cost. The first 1,000 units will be 2-3x. The first 10,000 will approach target cost. Volume 100,000 will be below target. Investors must understand this trajectory — early high costs are not failure; they’re the necessary prelude to competitive cost at scale.”

Document prepared for investor and operational planning purposes. All manufacturing cost estimates, timelines, and capacity projections are preliminary and subject to refinement through detailed process engineering and pilot production experience. Facility location decisions will be based on competitive site selection processes.